

**PUBLIC
EXHIBIT 52**

U.S. Patent No. 9,093,425 Domestic Industry Chart

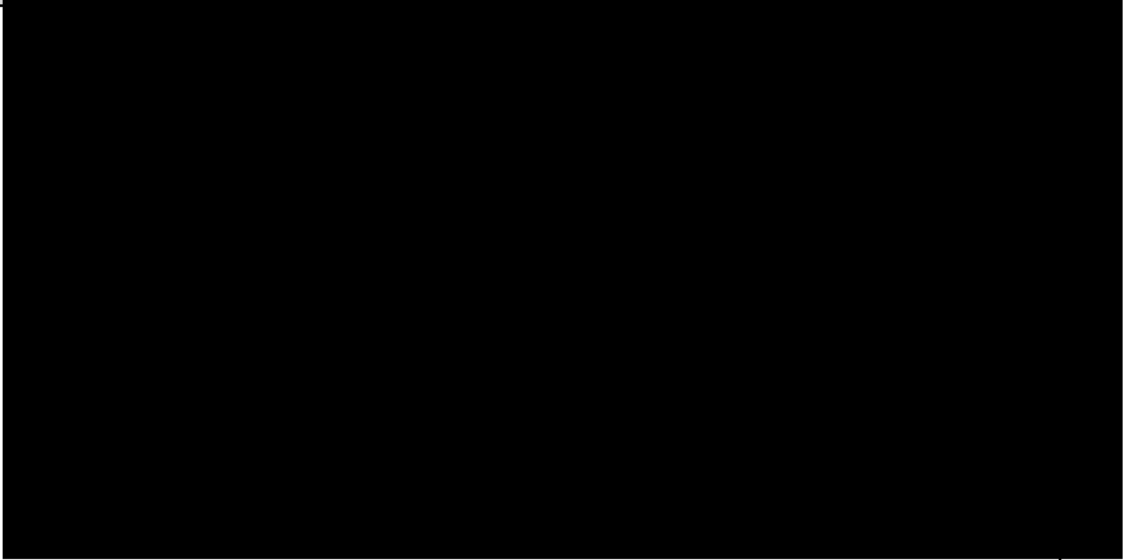
Title: Self-aligned liner formed on metal semiconductor alloy contacts

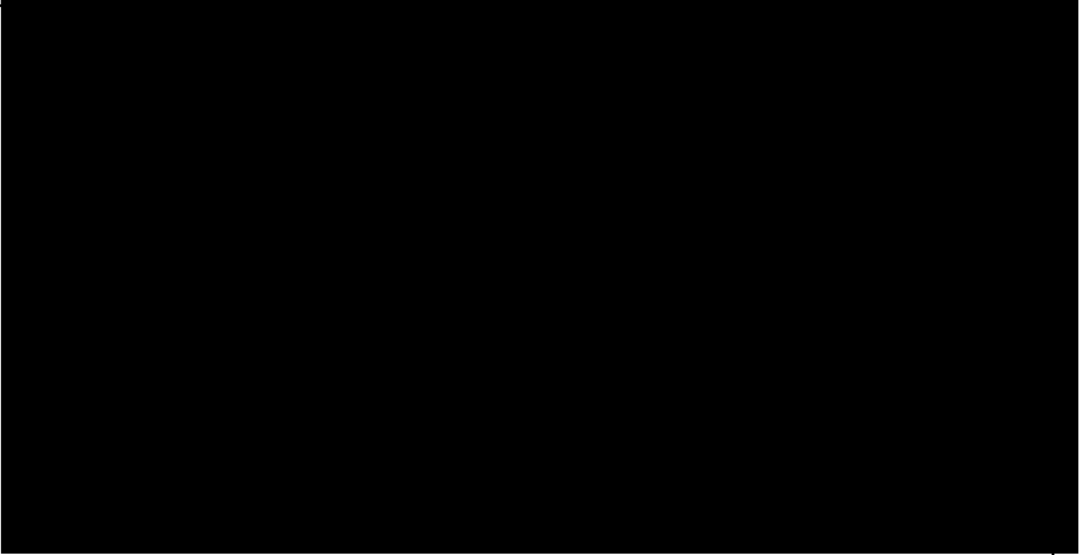
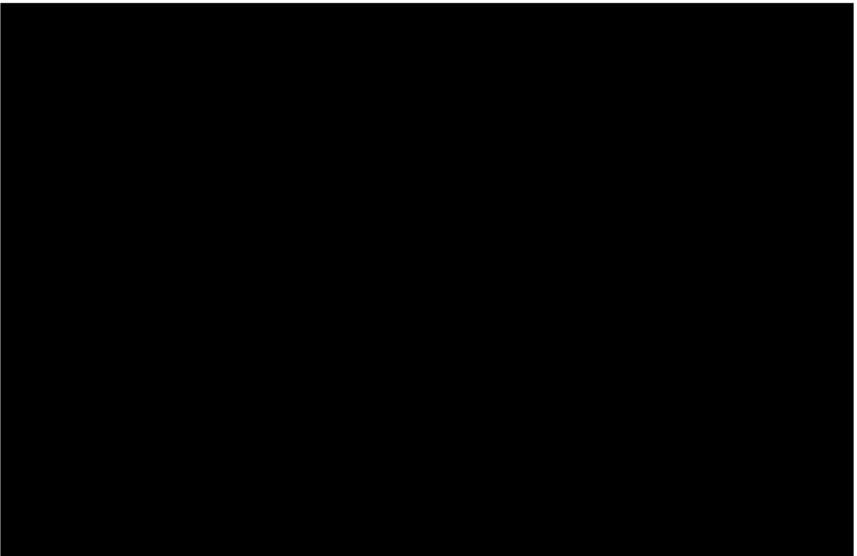
Inventors: Nicolas Breil, Christian Lavoie, Ahmet S. Ozcan, Kathryn T. Schonenberg, Jian Yu

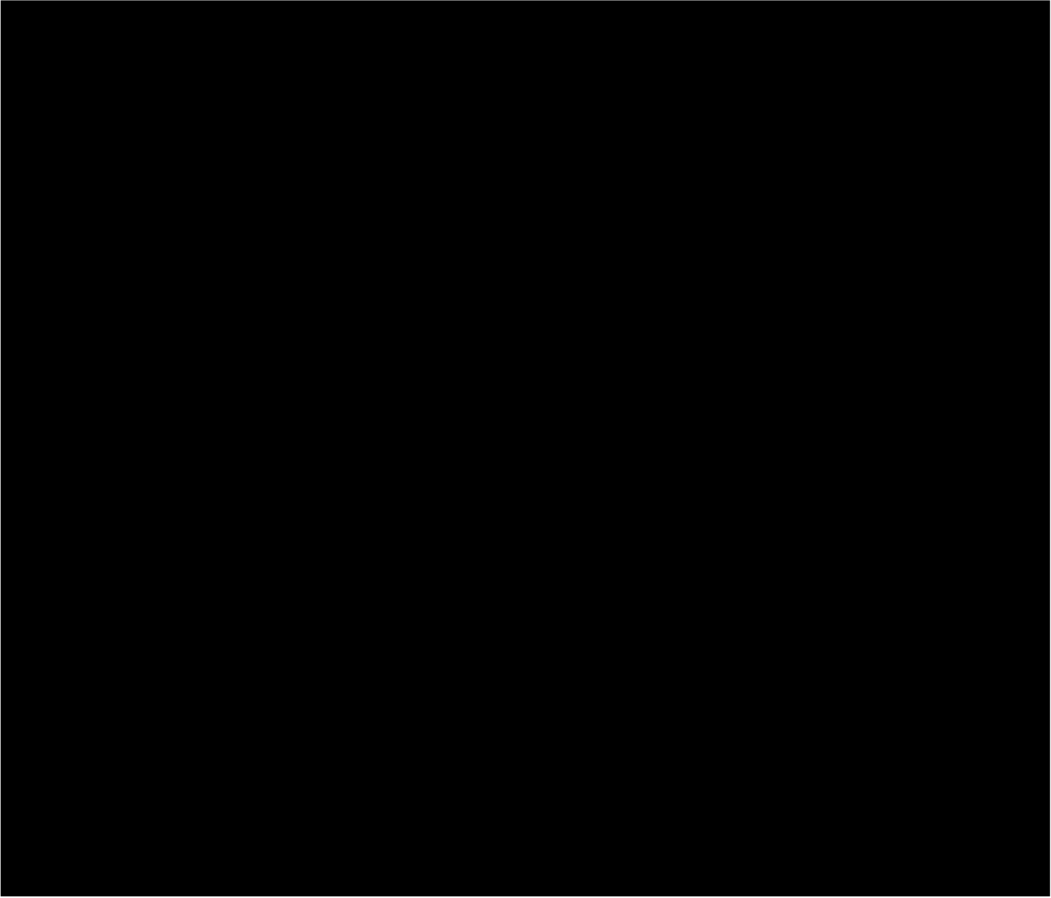
Abstract: Metal semiconductor alloy contacts are provided on each of a source region and a drain region which are present in a semiconductor substrate. A transition metal is then deposited on each of the metal semiconductor alloy contacts, and during the deposition of the transition metal, the deposited transition metal reacts preferably, but not necessarily always, in-situ with a portion of each the metal semiconductor alloy contacts forming a transition metal-metal semiconductor alloy liner atop each metal semiconductor alloy contact. Each transition metal-metal semiconductor alloy liner that is provided has outer edges that are vertically coincident with outer edges of each metal semiconductor alloy contact. The transition metal-metal semiconductor alloy liner is more etch resistant as compared to the underlying metal semiconductor alloy. As such, the transition metal-metal semiconductor alloy liner can serve as an effective etch stop layer during any subsequently performed etch process.

Claim	'425 Domestic Industry Products
[1pre] A semiconductor structure comprising:	To the extent the preamble is limiting, chips manufactured using GlobalFoundries' 130CBIC Process (hereinafter the "'425 Domestic Industry Products") ¹ comprise a semiconductor structure.

¹ GlobalFoundries reserves the right to identify different or additional '425 Domestic Industry Products and amend or supplement these charts after further investigation and discovery, claim construction, or other meriting circumstances.

Claim	'425 Domestic Industry Products
	
[1a] a semiconductor substrate having a source region and a drain region located within a semiconductor material portion of said semiconductor substrate, wherein said source region and said drain region are spaced apart by a channel region;	The '425 Domestic Industry Products comprise a semiconductor substrate having a source region and a drain region located within a semiconductor material portion of said semiconductor substrate, wherein said source region and said drain region are spaced apart by a channel region.

Claim	'425 Domestic Industry Products
	
[1b] a functional gate structure located above said channel region;	The '425 Domestic Industry Products comprise a functional gate structure located above said channel region. 

Claim	'425 Domestic Industry Products
[1c] a source-side material stack of, from bottom to top, a source-side metal semiconductor alloy portion and a source-side transition-metal metal semiconductor alloy portion located on one side of said functional gate structure and located atop said source region; and	The '425 Domestic Industry Products comprise a source-side material stack of, from bottom to top, a source-side metal semiconductor alloy portion and a source-side transition-metal metal semiconductor alloy portion located on one side of said functional gate structure and located atop said source region. For example, the images below shows the material stack for the source/drain region. The material analysis charts below show the composition of the source-side material stack. The green arrow pointing upwards in the figure below corresponds to the green arrow pointing from right to left in the following figure. 

Claim	'425 Domestic Industry Products
[1d] a drain-side material stack of, from bottom to top, a drain-side metal semiconductor alloy portion and a drain-side transition metal-metal semiconductor alloy portion located on another side of said functional gate structure and located atop said drain region.	The '425 Domestic Industry Products comprise a drain-side material stack of, from bottom to top, a drain-side metal semiconductor alloy portion and a drain-side transition metal-metal semiconductor alloy portion located on another side of said functional gate structure and located atop said drain region. <i>See</i> limitation [1c]. The source-side material stack and drain-side material stack in the '425 Domestic Industry Products are identical for given type of transistor, such that the description of limitation [1c] applies equally to the drain-side material stack (which is located on the other side of the functional gate structure).